Chatting in 64-bit

We got Randy Allen, Vice President of Design Engineering, Computation Products Group at AMD, to answer a few questions.

Digit (D): Which applications will benefit from the AMD64, and when?
Randy Allen (RA): Databases, etc., require 64-bit addressing, but high-end gaming and the 'cinematic experience', are also significant beneficiaries. Time-frames are something we have to work out, but it should be in 2004.

D: What have you done to make it perform so high at such low clock speeds? Are there any specific technologies to allow it to compete with Intel's Hypethreading?
RA: Naturally, when you look at clock speed, we have a frequency deficit against some of our competitors. So people ask, "How do you do it?" With the Athlon 64, we have made numerous improvements, though the two most significant are the integrated memory controller that reduces latency and the Hyper Transport bus. The Athlon 64 offers outstanding 32-bit performance, so it is better purely as a 32-bit processor anyway. But it's not simply a 32-bit processor—it allows you to move to 64-bit whenever you want. So I would turn that question around and ask why would anyone want to be limited to a 32 bit only system.

D: Will the benefits justify porting existing 32-bit applications to 64-bit today? And how will you ensure 64-bit driver support since 32-bit drivers won't work in 64-bit Windows?
RA: To give you an example, at the Opteron launch, IBM said that they ported each of their applications in a week—they were astonished at the simplicity. When we launch, we will have a host of chipsets, motherboards and systems from several vendors. And by the time 64-bit Windows comes around, we will be bringing hardware into our validation lab.

D: Will there be a Athlon 64 to FX upgrade path? (Note: The Athlon64 FX has a better integrated memory controller, and has a different socket configuration from the Athlon64.)
RA: Certainly in the future we would like to move to a common socket, but I wouldn't go as far as to say that there will be an upgrade path.

D: If Intel would like to use AMD64 architecture, would it be made available to them?
RA: Certainly! As a company, we are very active in licensing our technologies. Hyper Transport is a good example of a technology we have licensed. I think it will be very interesting to see Intel acknowledge that AMD64 is the standard.

tions for the Athlon 64 FX may be a problem.

In the real world video encoding tests that compare the system's ability to encode an MPEG-2 file to DivX, the Athlon outperformed the P4 by nearly 13 minutes. In the audio encoding tests, the Athlon lost out by 3 minutes. However, this small setback in the audio scores will not be a concern when working with multimedia applications. The POVRay scores rated the Athlon 64 FX lower than the AMD Athlon XP 3200+.

At the end of the day, the Athlon does give better performance on certain fronts—a conclusive verdict can only be passed after the benchmarking is a little more consolidated, and 64-bit OS' enter the scene.

The Athlon 64 FX is priced at $733, while the Pentium 4 3.2 GHz is available in the range of $600 to $700. To get maximum return over this investment, the Athlon 64 should excel in 32-bit mode since 64-bit applications will not be available for at least the next 18 months. Even when these applications are ported to 64-bit, the first ones will be professional applications like audio and video-editing tools. Intel is also releasing its P4 HT Extreme Edition that could give some stiff competition to the Athlon FX. AMD is also releasing a scaled-down version of the AMD Athlon 64 without the FX tag that is aimed at bringing affordable 64-bit computing to desktops. But until that happens, it makes sense to stick around with your old PC.

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